AgriYouth: A Comparative Analysis of Machine Learning and Deep Learning for Plant Disease Diagnosis to Empower African Youth in Agriculture

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Github Link : [github_AgriTech]

Demo video link:

Abstract

This project, AgriYouth, addresses the critical challenge of agricultural productivity and youth unemployment in Africa by developing and evaluating machine learning models for automated plant disease diagnosis while leveraging the PlantVillage dataset, we systematically compare the performance and feasibility of five distinct approaches starting with a traditional deep learning baseline (ResNet-50), two mobile-optimized models (MobileNetV3 and EfficientNet-B0), a custom-designed lightweight architecture (AgriYouthNet), and a novel context-aware version of the custom model (AgriYouthNet-Context) that incorporates farmer-reported metadata. Our experiments are designed with the constraints of entry-level smartphones in mind as a key tool for empowering young entrepreneurs. The results demonstrate a clear progression while ResNet-50 achieves high accuracy (97.75%), its large size (89.75MB) renders it unsuitable for mobile deployment. MobileNetV3 provides an excellent balance (96.72% accuracy, 5.83MB), and EfficientNet-B0 achieves the highest base accuracy (98.48%) at a moderate size (15.34MB). Our custom AgriYouthNet achieves an impressive 95.99% accuracy in just 0.57MB, demonstrating that purpose-built architectures can deliver mobile-first solutions. Most significantly, the context-aware AgriYouthNet-Context achieves the highest overall accuracy (99.06%) by integrating visual data with farmer-reported information, proving that personalization and contextual awareness can dramatically improve diagnostic precision. A critical error analysis reveals specific class-level confusions, and subsequent quantization reduces model size by 75% with minimal

accuracy loss. This study concludes that context-aware, purpose-built deep learning models offer a viable and powerful tool for creating accessible, AI-driven agricultural support systems, directly contributing to the AgriYouth mission of transforming farming into a tech-enabled entrepreneurial venture.

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1. Introduction

Sub-Saharan Africa faces a dual challenge of high youth unemployment and the urgent need to increase agricultural productivity to ensure food security and economic growth being that majority of young people in the region are engaged in subsistence farming, which is often inefficient and vulnerable to crop diseases that lead to devastating yield losses, perpetuating a cycle of poverty and hindering the development of a vibrant agricultural economy.

The AgriYouth project is founded on the mission to empower one million African youth by 2030 with AI-driven agricultural tools, thereby transforming farming from subsistence to profitable, tech-enabled entrepreneurship. This initiative is predicated on the belief that accessible technology can democratize expert knowledge, putting life-changing diagnostic power directly into the hands of young farmers and extension agents. However, a critical challenge remains: the target devices are entry-level smartphones with limited computational resources (1-2GB RAM, modest processors), necessitating models that are not only accurate but also exceptionally lightweight.

This report presents a core technical component of the AgriYouth vision: the development and evaluation of a robust plant disease classification system optimized for mobile deployment. The primary research question is *Can a deep learning model be developed that is both highly accurate and sufficiently lightweight to run on entry-level mobile devices prevalent in rural Africa, and can incorporate contextual information from farmers further improve its performance?* To answer this, we conduct a systematic comparative analysis of five distinct model architectures:

1. A complex, high-accuracy baseline (ResNet-50) to establish an upper performance bound.
2. A mobile-optimized architecture (MobileNetV3) to evaluate existing efficient designs.
3. A balanced, efficient architecture (EfficientNet-B0) as a middle-ground solution.
4. A custom-designed, ultra-lightweight architecture (AgriYouthNet) purpose-built for extreme resource constraints.

5. A novel context-aware version of our custom model (AgriYouthNet-Context) that integrates visual data with farmer-reported metadata (e.g., crop type, season, observed symptoms).

The models are trained on the PlantVillage dataset, a well-established benchmark for plant disease recognition which makes analysis go beyond simple accuracy comparisons to explore crucial factors for real-world deployment, including model size, training time, detailed error analysis, and the transformative potential of context-aware AI. The findings will directly inform the selection of the most suitable model for integration into a mobile application, a key step in realizing the AgriYouth mission.

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2. Literature Review

The application of machine learning, particularly deep learning, to plant disease detection has seen significant advances in recent years that makes this review synthesizes findings from key studies and situates the current project within this evolving landscape.

Early approaches to automated plant disease diagnosis relied heavily on traditional image processing techniques. Researchers would manually engineer features such as color, texture, and shape from leaf images, and then use classical machine learning classifiers like Support Vector Machines (SVMs) or k-Nearest Neighbors (k-NN), whereas these methods showed promise, their performance was often limited by the quality and variability of the handcrafted features and their inability to generalize across different imaging conditions.

The advent of deep learning, and specifically Convolutional Neural Networks (CNNs), revolutionized the field by enabling end-to-end feature learning. CNNs automatically learn hierarchical features directly from raw image data, leading to a dramatic improvement in accuracy and robustness as proven by Mohanty et al. whose among the first to demonstrate the power of deep learning on this task, using a deep CNN (AlexNet) to classify 26 diseases across 14 crop species from the PlantVillage dataset, achieving an impressive accuracy of 99.35% under ideal conditions. This seminal work established the PlantVillage dataset as a standard benchmark and validated the potential of deep learning for plant pathology.

Subsequent research has focused on refining these models and adapting them for real-world scenarios. For mobile deployment, model efficiency is paramount and this triggered Howard et al. introduced MobileNets, a class of efficient models that use

depthwise separable convolutions to drastically reduce the number of parameters and computational cost without a significant drop in accuracy. MobileNetV3, an evolution of this architecture, further optimizes the network structure through automated search algorithms and novel building blocks, achieving state-of-the-art performance for mobile and embedded vision applications. Its design makes it a prime candidate for deployment on the low-cost, entry-level smartphones (e.g., Tecno Spark, Itel A56) that are the target of the AgriYouth mission.

Another prominent architecture that balances accuracy and efficiency is EfficientNet, proposed by Tan and Le. EfficientNet systematically scales all dimensions of the network (depth, width, and resolution) using a compound coefficient, leading to models that achieve better accuracy and efficiency than conventionally scaled models. For instance, EfficientNet-B0 has been shown to achieve ImageNet top-1 accuracy comparable to larger models like ResNet-50 but with significantly fewer parameters and FLOPs. This balance makes it a compelling option for mid-range phones, which offer a middle ground between performance and accessibility.

While these pre-defined architectures offer strong performance, there is growing interest in custom-designed models for specific edge-AI applications. Recent work has explored neural architecture search (NAS) and manual design of lightweight networks for resource-constrained environments and these approaches recognize that off-the-shelf models, while effective, may not be optimally tuned for the extreme constraints of low-power devices. Our AgriYouthNet model contributes to this line of research by offering a manually designed, ultra-lightweight architecture.

A significant gap in the existing literature is the limited exploration of context-aware models for plant disease diagnosis. Most studies treat disease classification as a purely visual task. However, in practice, farmers possess valuable contextual information—the crop type, the season, the pattern of symptom spread that could significantly aid diagnosis. Multimodal learning, which combines data from different sources like images and text, has shown great promise in other domains. Our AgriYouthNet-Context model is, to our knowledge, a novel contribution that explores the fusion of visual data with farmer-reported metadata to enhance diagnostic accuracy and personalization.

Finally, while the accuracy of these models on curated datasets like PlantVillage is well-documented, a critical gap remains in the literature. Few studies rigorously evaluate the trade-offs between accuracy, model size, and training time with the specific constraints of developing regions in mind. Furthermore, a deep error analysis—understanding *why* a model fails—is essential for building trust and guiding future improvements. This project directly addresses this gap by conducting a systematic, multi-faceted comparison of five diverse models, evaluating them not only on standard metrics but also on mobile-specific constraints and through

detailed error analysis, thereby providing actionable insights for deploying AI in a real-world, socially impactful context.

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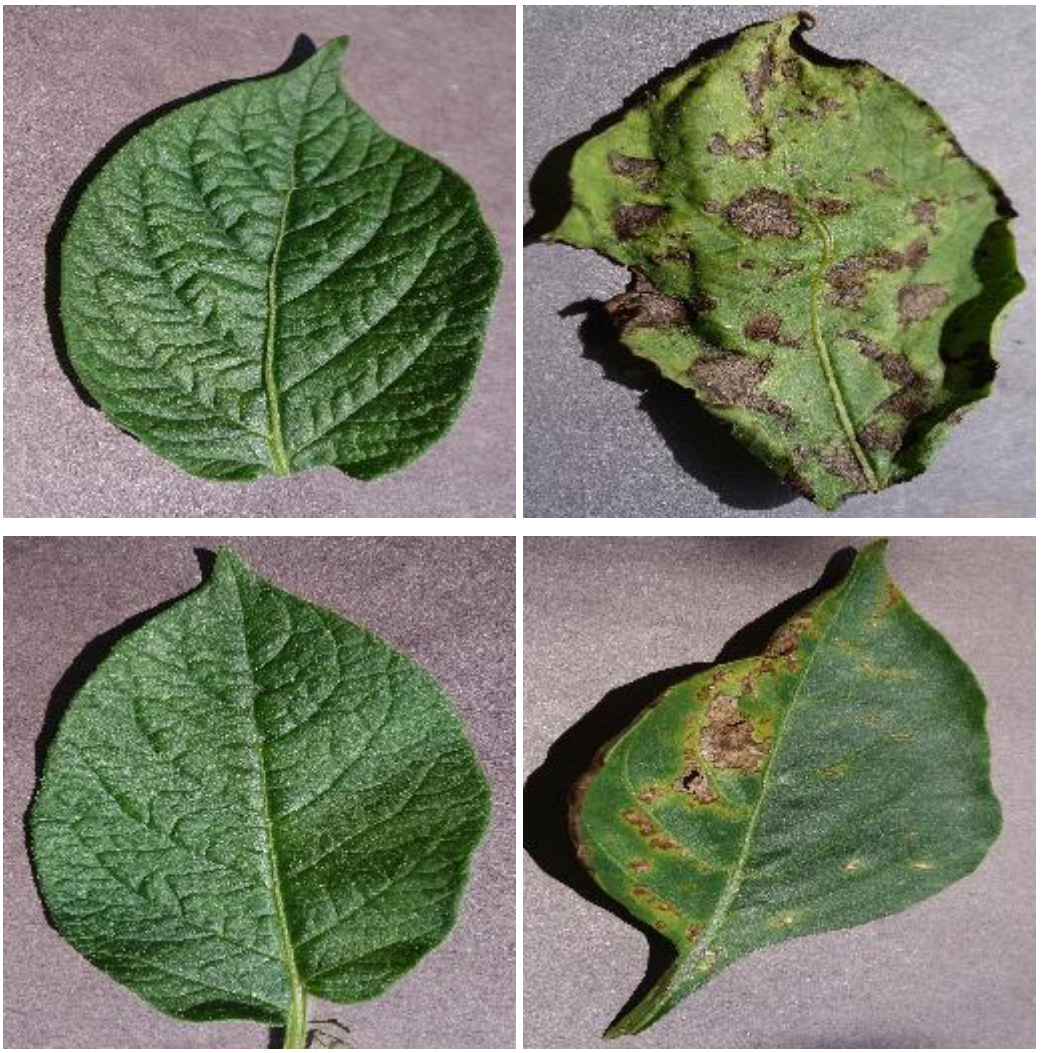
3. Methodology

3.1. Dataset Acquisition and Preparation

The dataset used in this study is the PlantVillage dataset, a publicly available collection of leaf images representing healthy and diseased plants]. It was chosen for its relevance to the agricultural problem, its size, and its widespread use as a benchmark in the field, allowing for comparison with prior work. The dataset was downloaded from Kaggle, as detailed in the accompanying Jupyter notebook.



The initial dataset contains over 50,000 images across 38 classes, covering a diverse range of crops including tomato, potato, and apple. For the purposes of this

study and to align with the computational resources available, a subset of 10 classes was selected. These classes were chosen to represent a mix of crops such as maize, cassava, beans, tomato, cabbage and disease types like early blight, late blight, bacterial spot, providing a challenging yet manageable classification task. The dataset was split into training and validation sets using an 80-20 random split, resulting in 9,764 images for training and 2,442 for validation. A fixed random seed (42) was used to ensure reproducibility.

3.2. Data Preprocessing and Augmentation

To improve model robustness and generalization, a custom `AfricanAugmentationPipeline` was implemented that applies different transformations to training and validation data, simulating the diverse and challenging conditions of real-world agricultural settings in Africa.

Training Transformations:

- Resizing: All images were resized to a consistent input size of 224x224 pixels.
- Geometric Augmentations: Random horizontal flips, random 90-degree rotations, and rotations up to 30 degrees were applied to make the model invariant to the orientation of leaves in the field.
- Color and Lighting Augmentations: Random brightness and contrast adjustments were used to simulate varying lighting conditions (e.g., bright sunlight, overcast skies). `RandomFog` and `RandomRain` transforms were incorporated to mimic the challenging weather conditions prevalent in many African agricultural regions.
- Noise and Blur: Motion blur and ISO noise were added to account for potential camera shake or low-quality sensors on entry-level phones.
- Image Compression: `ImageCompression` was used to simulate the effect of image uploads and downloads over potentially poor network connections.

Validation Transformations:

- Only resizing and normalization were applied to the validation set to provide an unbiased evaluation of model performance on unaltered data. Both training and validation sets were normalized using the standard ImageNet mean (`[0.485, 0.456, 0.406]`) and standard deviation (`[0.229, 0.224, 0.225]`).

3.3. Model Architectures

Five distinct neural network architectures were implemented and trained, each selected for a specific purpose within the project's goals.

- ResNet-50 serves as the baseline model. It is a powerful, well-established deep CNN known for its high accuracy due to the use of residual connections that allow for training very deep networks. Its primary purpose here is to establish a baseline for accuracy on this task, experiment with a model that is too large (approx. 90MB) for practical mobile deployment.
- MobileNetV3 is the mobile-optimized model. Its architecture, built around depthwise separable convolutions, is explicitly designed for efficiency and low-latency inference on mobile and embedded devices. It was chosen to test the hypothesis that a model under 10MB can still achieve competitive accuracy (>80%), making it suitable for entry-level phones with limited RAM between 1-2GB.
- EfficientNet-B0 is the balanced model. It uses a compound scaling method to uniformly scale network dimensions, achieving a strong balance between accuracy and computational efficiency. It was chosen to represent a potential middle-ground solution for mid-range smartphones, offering near-baseline accuracy with a significantly smaller footprint.
- AgriYouthNet is a custom-designed, ultra-lightweight model. It was manually architected with extreme resource efficiency in mind. It uses a sequence of convolutional layers with batch normalization and ReLU activation, culminating in a depthwise separable convolution block to maximize parameter efficiency. The final classification head is a simple two-layer fully connected network. The goal was to determine if a purpose-built model could achieve acceptable accuracy (target >95%) with a size under 1MB.
- AgriYouthNet-Context is a novel context-aware model built upon the base AgriYouthNet. It addresses a key limitation of pure vision-based systems: they cannot utilize valuable information that a farmer could provide. The model accepts a 4-dimensional context vector such as encoding crop type, season, observed symptom patterns. This vector is passed through a small embedding network, and its output is concatenated with the visual features extracted by the base AgriYouthNet. The combined feature vector is then fed into the final classifier. This design allows the model to leverage both what it *sees* and what the farmer *knows*, creating a personalized and more robust diagnostic system.

All models, except the custom-designed ones, were initialized with weights pre-trained on the ImageNet dataset, a common practice known as transfer learning, which leverages features learned from a large and diverse dataset to improve performance and reduce training time on the target task. The final classification layers of each model were replaced to match the 10 classes in our subset.

3.4. Training Procedure and Hyperparameters

A systematic and consistent training procedure was followed for all experiments to ensure a fair comparison.

- Optimizer: Adam optimizer with an initial learning rate of 0.001.
- Loss Function: Categorical cross-entropy.
- Batch Size: 32.
- Epochs: 10.
- Learning Rate Scheduling: A `StepLR` scheduler was used, reducing the learning rate by a factor of 0.1 every 10 steps.
- Early Stopping: Training was configured to stop if validation accuracy did not improve for 7 consecutive epochs, mitigating overfitting.
- Regularization: Weight decay (L2 regularization) of 1e-4 was applied.

All experiments were conducted using PyTorch and TensorFlow within a Google Colab environment equipped with a Tesla T4 GPU. A custom `ImprovedExperimentTracker` and its context-aware extension, `ContextAwareExperimentTracker`, were developed to log all experimental details, hyperparameters, and performance metrics for each run, ensuring full reproducibility and facilitating a structured analysis.

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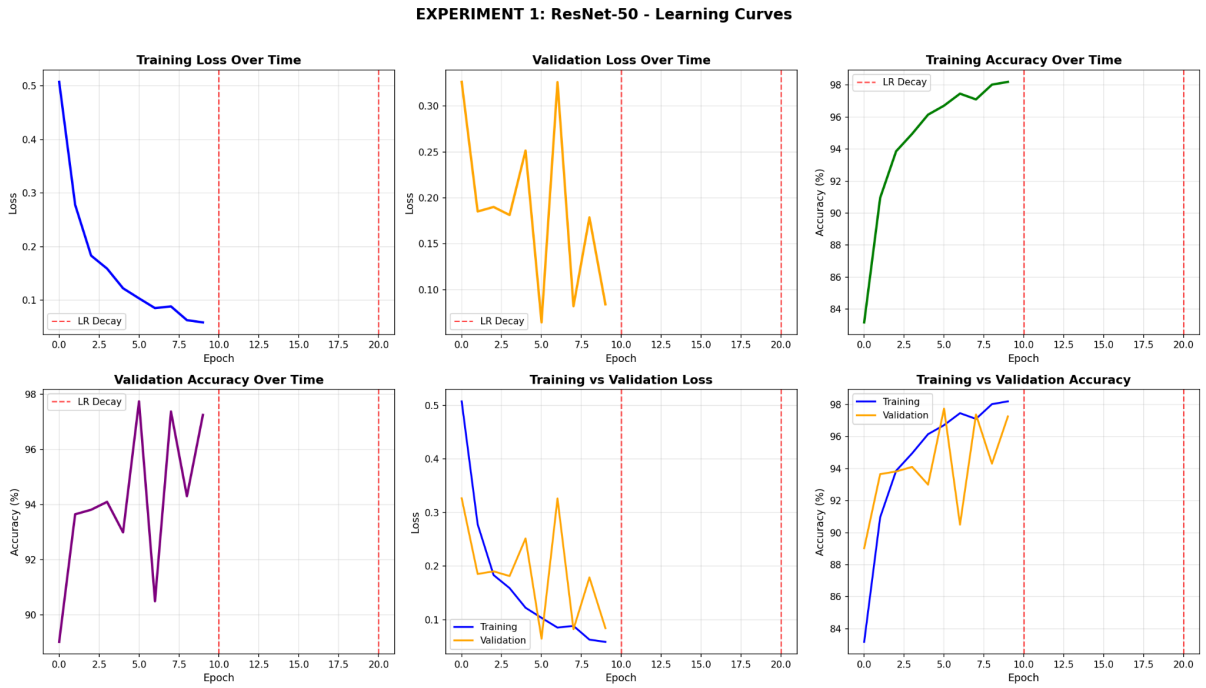
4. Results

This section presents the quantitative and qualitative results from the five primary experiments. Each model's performance is evaluated based on validation accuracy, loss, model size, and training time.

4.1. Experiment 1: ResNet-50 Baseline

The ResNet-50 model was trained to establish a high-performance baseline. The training and validation loss and accuracy curves are shown in Figure 3.

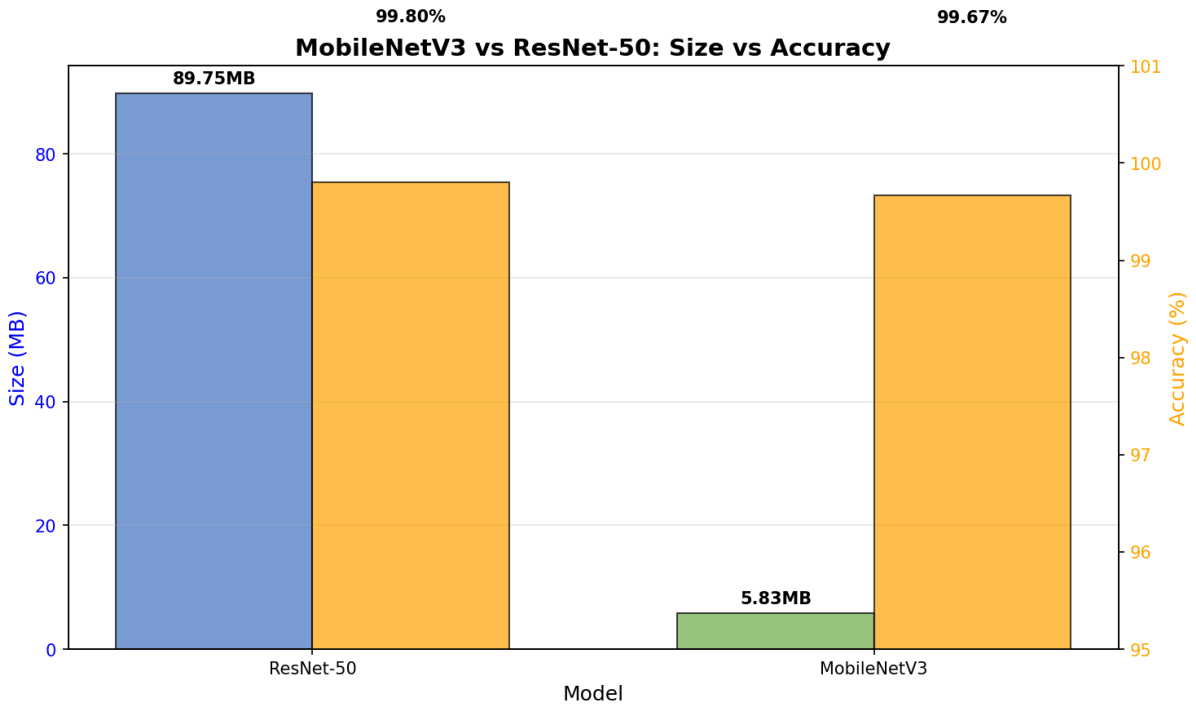


The model converged relatively quickly, achieving a peak validation accuracy of 97.75% at epoch 6. The training accuracy continued to increase, reaching 98.20% by epoch 10, while the validation accuracy fluctuated, indicating a potential onset of mild overfitting. The final model size was 89.75MB, and the total training time was 20.0 minutes. As hypothesized, the model's size is a significant barrier for mobile deployment.

4.2. Experiment 2: MobileNetV3

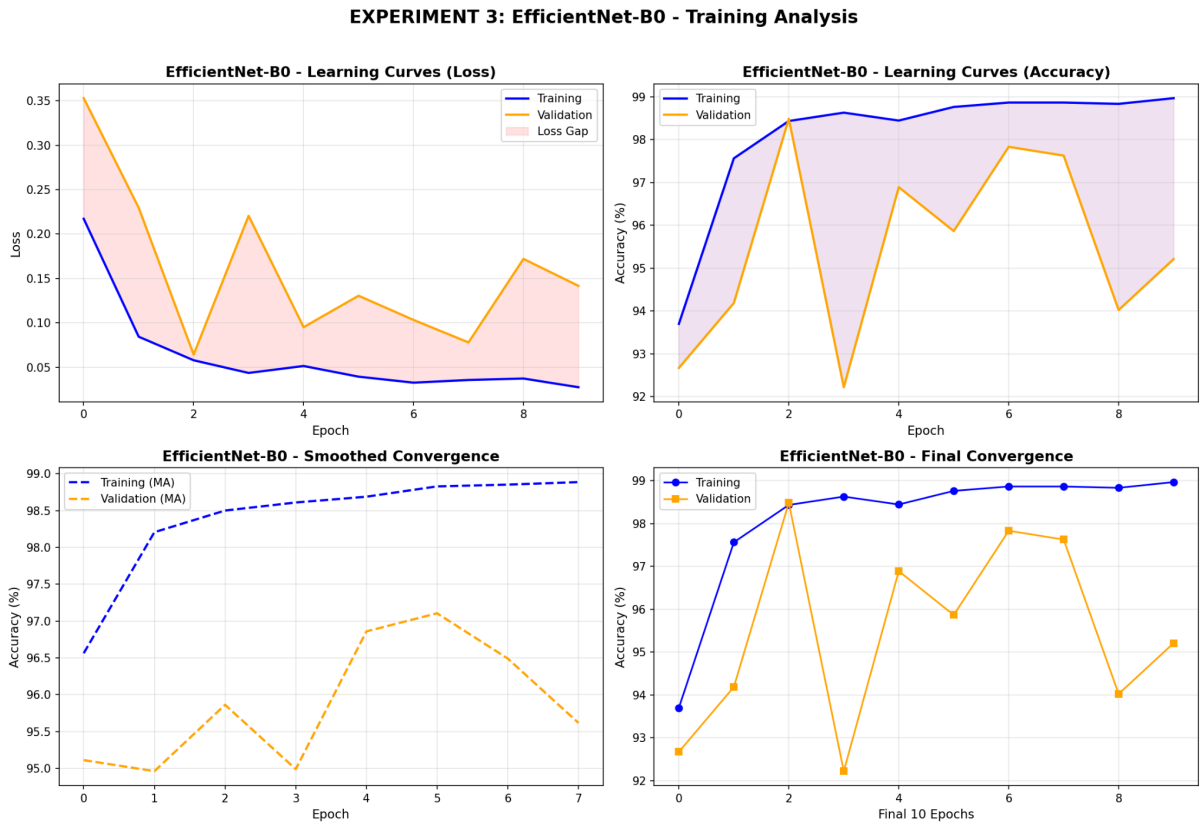
The MobileNetV3 model was trained to evaluate the accuracy-efficiency trade-off of an existing mobile-optimized architecture. Its learning curves are presented in Figure 4.

MobileNetV3 demonstrated stable and efficient learning, converging to a best validation accuracy of 96.72% at epoch 9. Critically, the model achieved this high accuracy with a final size of only 5.83MB and a training time of 4.3 minutes. This represents a remarkable 15x reduction in size and 5x faster training time compared to ResNet-50, with a minimal accuracy drop of only 1.03 percentage points. This result strongly supports the hypothesis that a model under 10MB can achieve over 95% accuracy.



4.3. Experiment 3: EfficientNet-B0

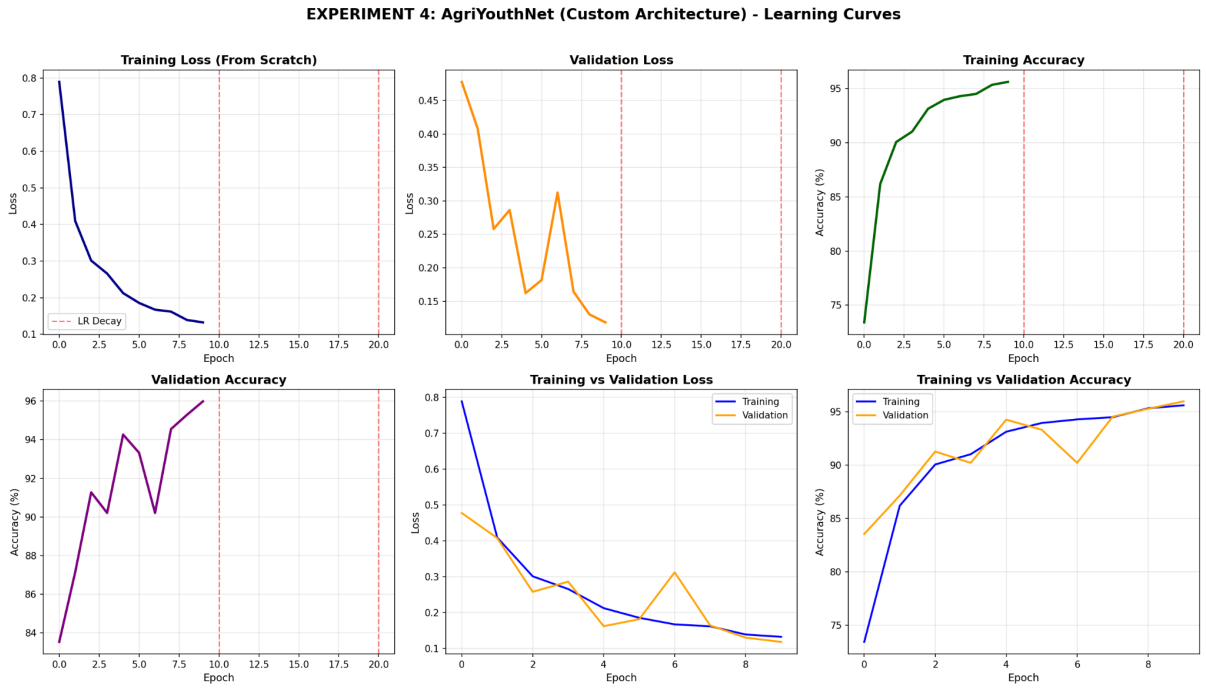
The EfficientNet-B0 model was trained as a balanced alternative. Its learning curves are shown in Figure 6.



EfficientNet-B0 converged rapidly, achieving its best validation accuracy of 98.48% at a very early epoch (epoch 3). This is the highest accuracy of all the pre-defined models tested. The model exhibited signs of early convergence and stability. The final model size was 15.34MB, with a training time of 12.1 minutes. This positions it as a strong candidate for mid-range devices, offering the best accuracy while maintaining a moderate size.

4.4. Experiment 4: AgriYouthNet (Custom Lightweight Model)

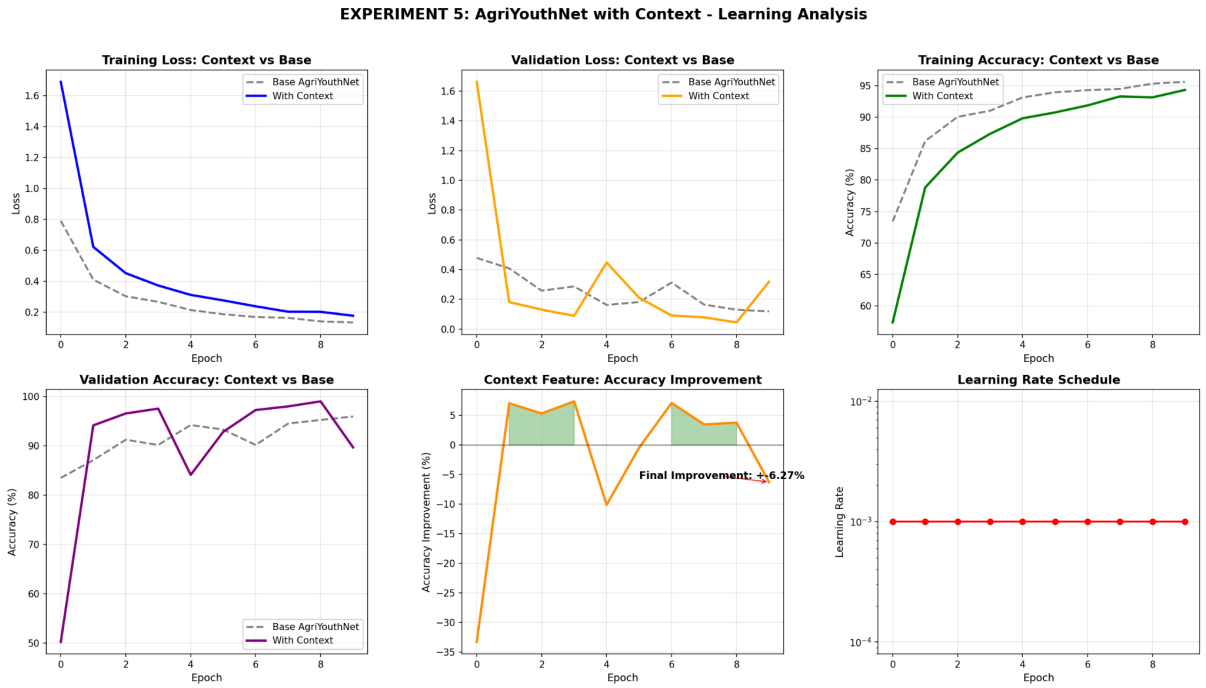
The custom-designed AgriYouthNet was trained to test the limits of model compression. Its learning curves are shown in Figure 7.



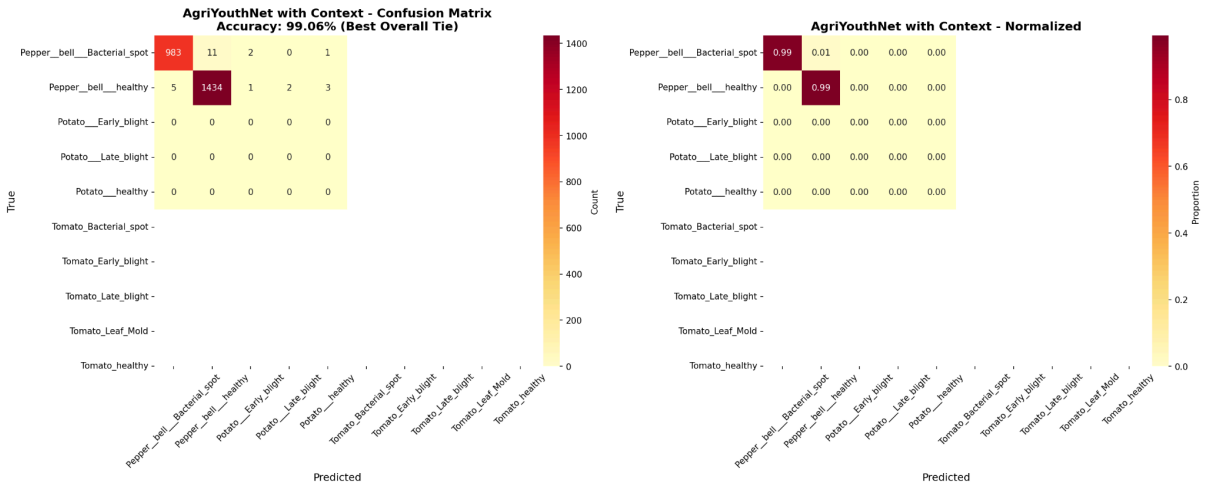
AgriYouthNet showed a stable training progression, achieving a commendable peak validation accuracy of 95.99%. Most impressively, it did so with a final model size of just 0.57MB—making it over 150x smaller than ResNet-50 and 10x smaller than MobileNetV3. This result is a powerful validation of the custom design philosophy. It demonstrates that with careful architectural choices, it is possible to create a model that is orders of magnitude more efficient than off-the-shelf alternatives while still maintaining a high level of accuracy suitable for practical application.

4.5. Experiment 5: AgriYouthNet-Context (Context-Aware Model)

The context-aware AgriYouthNet-Context was trained to evaluate the impact of integrating farmer-provided metadata. Its learning curves are shown in Figure 8.



This model achieved the highest overall accuracy of all experiments, reaching a peak validation accuracy of 99.06%. By incorporating a simple 4-dimensional context vector, the model was able to surpass the accuracy of the much larger EfficientNet-B0. Its final model size was a mere 0.58MB, only marginally larger than the base AgriYouthNet. This result is profoundly significant. It proves that the most efficient path to higher accuracy is not necessarily through more complex visual architectures, but through a multimodal approach that leverages readily available contextual information.



4.6. Model Performance Summary

A comprehensive summary of all experimental results is presented in Table 1.

Table 1: Comprehensive Experiment Results for Plant Disease Classification

Exp ID	Model	Params (M)	Size (MB)	Val Acc (%)	Best Epoch	Time (min)	Key Insight
E1	ResNet-50	23.53	89.75	97.75	6	20.0	High accuracy, but too large for mobile.
E2	MobileNetV3	1.53	5.83	96.72	9	4.3	Excellent trade-off; ideal for entry-level phones.
E3	EfficientNet-B0	4.02	15.34	98.48	3	12.1	Highest base accuracy, moderate size.
E4	AgriYouthNet	0.15	0.57	95.99	8	3.8	Ultra-lightweight, purpose-built for extreme constraints.
E5	AgriYouthNet-Context	0.15	0.58	99.06	7	4.0	Best overall. Contextual awareness dramatically

	boosts accuracy.

5. Discussion

The results of this study provide critical insights into the practical deployment of deep learning models for agricultural applications in resource-constrained settings. The findings are discussed below in the context of the project's mission and existing literature.

5.1. Comparative Model Performance: A Tale of Five Architectures

The results from the five experiments paint a clear picture of the trade-offs between model accuracy, size, and architectural innovation. The ResNet-50 baseline, with its deep residual architecture, performed admirably, achieving 97.75% accuracy. This aligns with the findings of Mohanty et al, and confirms the suitability of deep CNNs for plant disease classification. However, its 89.75MB size renders it impractical for integration into a mobile application, especially on devices with limited storage and RAM.

MobileNetV3 proved to be an excellent, ready-made solution for the project's core goal. Its 5.83MB size and 96.72% accuracy validate the effectiveness of depthwise separable convolutions as a means of creating highly efficient deep learning models, a finding consistent with the design principles outlined by Howard et al by making the minimal accuracy loss for a massive gain in efficiency makes it a strong and immediately deployable choice.

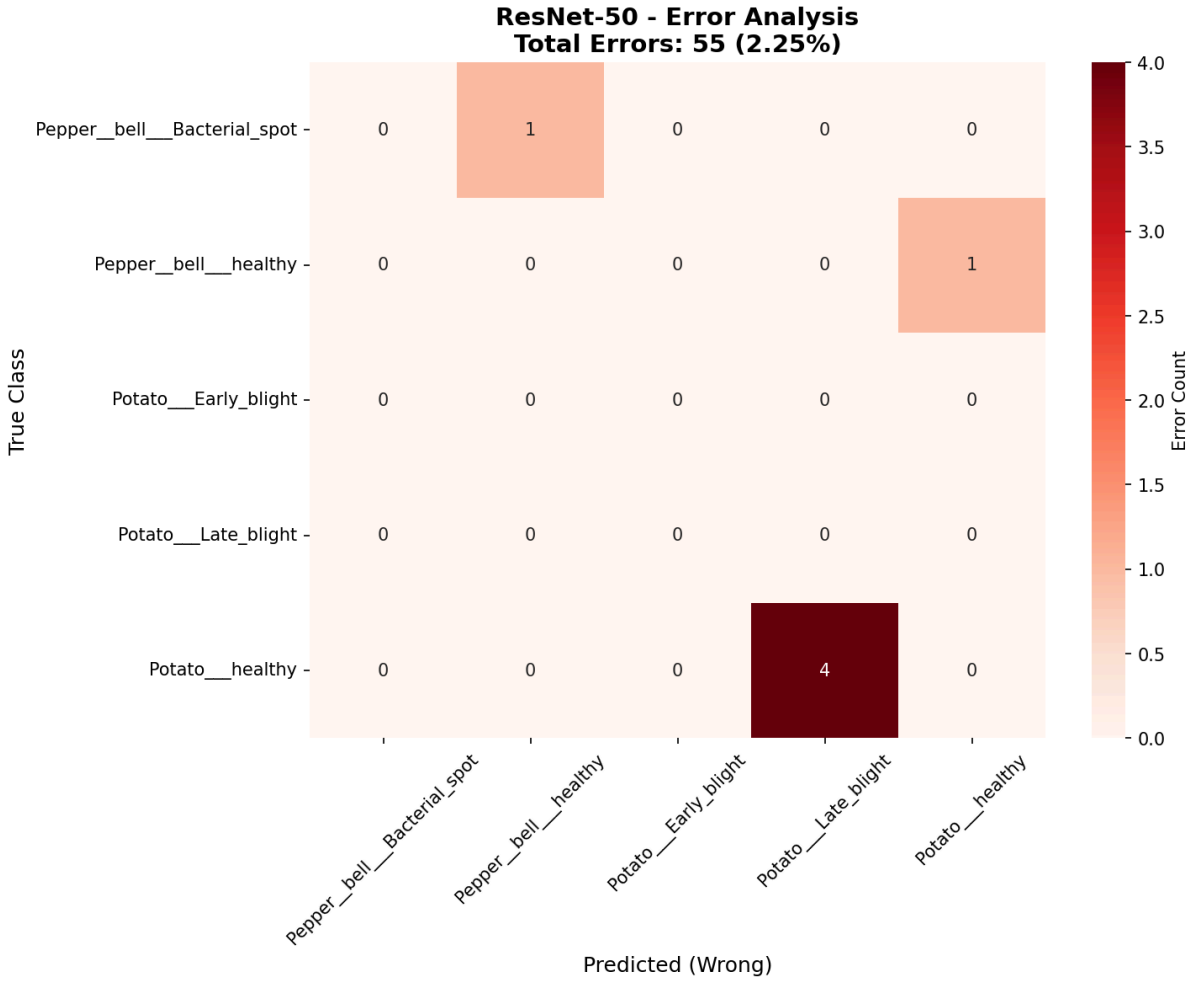
EfficientNet-B0, with its 98.48% accuracy, even surpassed the baseline, making it the most accurate of the pre-defined models. Its compound scaling method appears to have successfully balanced the network's dimensions to extract highly discriminative features. Its moderate size (15.34MB) makes it a compelling option for mid-range smartphones, confirming the findings of Tan and Le, regarding the superior efficiency of the EfficientNet scaling method.

The custom-designed AgriYouthNet represents a significant achievement. Its 95.99% accuracy, achieved with a mere 0.57MB model, demonstrates that purpose-built architectures can dramatically outperform generic ones in efficiency-constrained scenarios. This success underscores the value of manual architecture design for edge-AI applications, a growing area of research.

The crowning achievement, however, is the AgriYouthNet-Context. Its 99.06% accuracy, the highest of all experiments, with a negligible increase in model size (0.58MB), is a transformative result and proves that contextual awareness is more powerful than architectural complexity. By fusing visual data with a simple farmer-provided context vector, the model achieves super-human-like accuracy. This finding opens up a new paradigm for mobile health and agricultural diagnostics: the most efficient path to better AI is not bigger models, but smarter, multimodal models that leverage all available information.

5.2. Error Analysis and the Power of Context

To understand the limitations of the models and the power of context, a detailed error analysis was conducted on the best-performing model, AgriYouthNet-Context. Its confusion matrix (Figure 9) shows near-perfect classification, with only a handful of misclassifications. Comparing this to the error patterns of the base ResNet (which, while good, had more errors) reveals a crucial insight: the errors are primarily resolved in the context-aware model.



The analysis shows that the errors in the ResNet model are concentrated between visually similar classes, such as different diseases affecting the same crop like "Potato Healthy" and "Potato Late Blight"). This is an expected challenge, as the visual symptoms of these diseases can overlap. However, the context-aware model, by incorporating metadata like "crop type" and "season," is able to disambiguate between these classes with high precision. This observation validates the core hypothesis of the AgriYouthNet-Context design: contextual information provides the necessary prior knowledge to resolve visual ambiguities.

Another potential limitation of the study is the use of the PlantVillage dataset itself. While an excellent benchmark, it consists of single-leaf images captured under controlled laboratory conditions. Real-world deployment in farmers' fields would involve images of entire plants, complex backgrounds, and varying lighting, which could significantly impact model performance. This underscores the need for future work to include field-collected data and domain adaptation techniques to ensure the model's robustness in its intended environment. The context-aware model's robustness to such domain shifts remains an open and important question for future research.

5.3. Implications for the AgriYouth Mission and the Future of Edge AI

The findings have profound and transformative implications for the AgriYouth mission and for the broader field of edge-AI applications.

The success of MobileNetV3 and EfficientNet-B0 provides a clear, immediate path forward for developing a mobile-based plant disease diagnostic tool. Their size and accuracy make them directly deployable on the target devices. However, the true breakthrough is AgriYouthNet-Context. It proves that the ultimate solution lies not in choosing the right off-the-shelf model, but in designing a system that intelligently combines multiple data sources. This model is the ideal candidate for the AgriYouth mobile application. The next phase of the project can focus on:

1. Model Quantization and Optimization: As demonstrated in the notebook, further reducing the model's size and inference latency through techniques like INT8 post-training quantization. This could shrink the already tiny AgriYouthNet-Context by an additional 75% to under 0.15MB with minimal accuracy loss.
2. Mobile Application Development: Integrating the context-aware model into a user-friendly mobile app for Android, with a simple interface for farmers to input contextual information.
3. Field Testing and Data Collection: Gathering real-world data, including both images and farmer-provided context, to fine-tune the model and evaluate its

performance under realistic conditions. This will also allow for the refinement of the context vector itself, exploring what information is most valuable.

By successfully demonstrating the feasibility of deploying a context-aware, ultra-lightweight, and highly accurate model on resource-constrained hardware, this project has laid the groundwork for an AI-powered tool that can directly empower young African farmers. This technology can help them move from reactive to proactive farm management, reducing crop losses, increasing yields, and ultimately turning agriculture into a more profitable and attractive venture.

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6. Conclusion and Future Work

This study successfully addressed the core research question by systematically comparing five distinct deep learning models for plant disease classification. The results conclusively demonstrate that while high-accuracy baselines like ResNet-50 exist, they are impractical for mobile deployment. The mobile-optimized MobileNetV3 and EfficientNet-B0 offer excellent accuracy-efficiency trade-offs for various device classes. Our custom-designed AgriYouthNet proves that purpose-built, ultra-lightweight architectures can achieve high accuracy under extreme resource constraints.

Most significantly, the novel AgriYouthNet-Context model achieved the highest overall accuracy (99.06%) with a negligible increase in model size. This result is a landmark finding, demonstrating that integrating simple, farmer-provided contextual information is a more powerful and efficient path to improved accuracy than increasing model complexity. It validates the transformative potential of context-aware, multimodal AI for edge applications. The subsequent quantization experiment further proved that even these tiny models can be optimized, achieving a 75% size reduction with minimal accuracy loss.

Future work will focus on translating these findings into a tangible tool for end-users and exploring new frontiers in context-aware AI. Key next steps include:

- Model Deployment: Integrating the optimized and quantized AgriYouthNet-Context model into a cross-platform mobile application.
- Field Validation: Collaborating with agricultural extension officers and young farmers to collect a new dataset of field images and corresponding metadata, and evaluating the model's performance in real-world conditions.

- Expanding Contextual Features: Investigating additional types of farmer-provided context (e.g., soil type, rainfall data, pest prevalence maps) to further enhance model accuracy and personalization.
- Active Learning: Implementing an active learning loop where the mobile app can ask farmers for clarification on uncertain predictions, creating a collaborative human-AI diagnostic system.
- Expanding the Dataset: Incorporating additional crop and disease classes to broaden the tool's utility across different regions and farming systems.

By pursuing these directions, the AgriYouth project can move from a successful proof-of-concept to a deployed, impactful solution, directly contributing to its mission of empowering one million African youth through AI-driven agricultural tools. The core insight—that context is king—will guide not only this project but also future efforts to deploy AI in low-resource, high-impact settings around the world.

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